higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

ENGINEERING SYSTEMS
NQF LEVEL 2

27 FEBRUARY 2019

This marking guideline consists of 6 pages.

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QUESTION 1

1.1
1.1.1 backlash
1.1.2 cam
1.1.3 swelling
1.1.4 lubrication
1.1.5 maintenance

(5 × 1) (5)

1.2
1.2.1 True
1.2.2 True
1.2.3 False
1.2.4 True
1.2.5 False

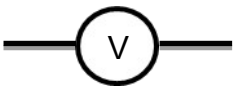
(5 × 1) (5)

1.3
1.3.1 Power
1.3.2 Lockout/Tagout/Isolate
1.3.3 Conductor
1.3.4 Actuator/Cylinder
1.3.5 Resistor

(5 × 1) (5)
[15]

QUESTION 2

2.1
2.1.1 

2.1.2 

2.1.3 

2.1.4 

2.1.5 

(5 × 1) (5)

2.2
2.2.1 Oil filter
2.2.2 Idler sprocket
2.2.3 Air receiver/Tank
2.2.4 Transformer
2.2.5 Control valve

(5 × 1) (5)

2.3 **Conductors**

- Copper
- Silver
- Tungsten
- Gold
- Aluminium (Any TWO)

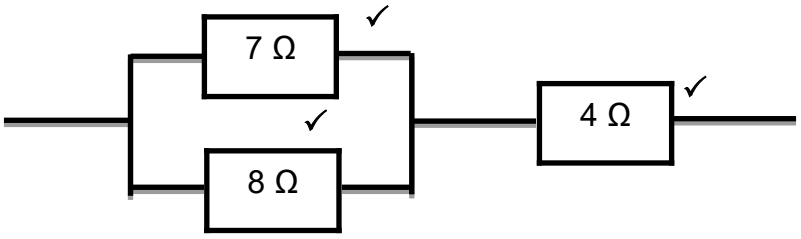
Insulators

- Rubber
- Plastic
- Wood
- Glass (Any TWO)

(2 × 2) (4)

2.4
- A transistor can switch current on and off.
- A transistor can make current stronger/amplify a current. (2)

2.5 2.5.1



(3)

2.5.2 $\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2}$

$\frac{1}{R_p} = \frac{1}{7} + \frac{1}{8}$✓

$\frac{1}{R_p} = \frac{15}{56}$✓

$R_p = \frac{56}{15}$✓

$R_p = 3{,}733\ \Omega$✓ (4)

2.5.3 $R_T = R_p + R_S$

$R_T = 3{,}733 + 4$✓

$R_T = 7{,}733\Omega$✓ (2)

[25]

QUESTION 3

3.1 3.1.1

A Shaft/Brake Drum
B Brake/Brake shoe
C Solenoid
D Torque adjuster

(4 × 1) (4)

3.1.2 When the electrical supply is cut, the brake is activated immediately and prevents rotation of the shaft under load, preventing the load from falling. This prevents injury to people or damage to equipment. (2)

3.1.3
- Overhead cranes
- Lifts
- Hoists (Any 1 × 1) (1)

3.2 Belt drives require friction between the pulley and the belt✓ to operate efficiently.✓ If the belt becomes contaminated with oil, slip occurs between the belt and the pulley.✓ Slip results in premature wear on the belt.✓
(Any relevant answer) (4)

3.3 Hydraulic systems use hydraulic or fluid pressure for work to be done, while pneumatic systems use compressed air to do work. (Any relevant answer) (2)

3.4 Gear drives have teeth that mesh eliminating the chance of slip and thus increase efficiency.✓✓ Belt drives rely on friction between the belt and the pulley to transmit power. Slip occurs between the pulley and the belt, reducing efficiency of the system.✓✓ (4)

3.5
- Filter with a water trap
- Pressure-reducing valve
- Lubricator (3)

[20]

QUESTION 4

4.1 4.1.1
- Gear X
- Gear Z (2)

4.1.2
- Input gear – gear W
- Output gear – gear Z (2)

4.1.3 $N_W x T_W = N_X x T_X$

$1800x50 = N_X x200$ ✓

$N_X = \frac{90000}{200}$

$N_X = 450\ r/min$ ✓ (2)

4.1.4 $N_Y x T_Y = N_Z x T_Z$

$450x40 = N_Z x160$ ✓

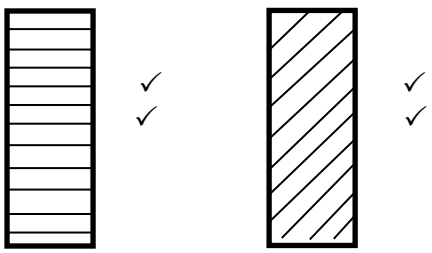
$N_Z = \frac{18000}{160}$

$N_Z = 112{,}5\ r/min$ ✓ (2)

4.2
- Processor
- Inputs and outputs
- Power supply
- Rack and chassis
- Communication interfaces
- Programming tools (Any 5 × 1) (5)

4.3
- Electromagnetic brakes
- Hydraulic brakes
- Pneumatic brakes
- Mechanical brakes (Any 3 × 1) (3)

4.4 Spur gear Helical gear



✓ ✓ ✓ ✓ (2 + 2) (4)

[20]

QUESTION 5

5.1
- Temperature gauges
- Pressure gauges
- Switches
- Thermostats
- Proximity switches (Any 3 × 1) (3)

5.2
- Inspect the machine to ensure all guards and other safety devices are in place.
- Notify all employees affected that the machine will be returned to service.
- Check the area to ensure that turning on the energy source and starting up the machine will not endanger employees.
- Double check if all bolts were tightened.
- Double check that no maintenance tools and equipment have been left in the machines.
- Remove the lockout devices.
- Turn on the energy of the machine.
- Ensure the rotational machinery does not overheat or make noises and that it is performing to its full capacity. (Any 5 × 1) (5)

5.3
- Check for cracks.
- Check for swelling.
- Check for abrasions.
- Check for excessive stretching.
- Check for damaged pulleys and misalignment. (Any 3 × 1) (3)

5.4
- Greater safety
- Smaller risk
- Higher quality of goods produced
- Lower operating costs
- Longer lifespan of machinery
- Higher end value of goods (Any 3 × 1) (3)

5.5
- Changes the speed of motion
- Changes the direction of motion
- Changes the direction of rotation
- Increases or decreases the force of the machine
- Converts to a different type of motion (Any 3 × 1) (3)

5.6
- Helps to compile a history on the performance of a particular machine
- Establishes a scheduled maintenance task
- Helps to determine if a machine should be replaced (3)

[20]

TOTAL 100

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